Representable functions satisfy a Q-provable uniqueness formula

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the 'if' direction is the author's own pointer to definition clause rep:a (a definitional restatement, not a delegated proof); the inconsistency coda cites the soundness theorem via the author's tagrefs marker, delivered verbatim

Editorial note. The label lem:rep-q names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Definition of representability. Same unit, representability definition (rep:a / rep:b clauses). The 'if' direction is exactly clause rep:a of the definition; rep:b supplies the uniqueness universal used in the 'only if' half.
- Q proves distinct numerals inequal. Same unit, \ollabel{lem:q-proves-neq} (statement lines 476-479). Provides the contradictory inequation Q proves num l != num m in the only-if argument.

Source reference. Original lemma lem-q-proves-neq: sources/olp-incompleteness-representability-in-q/pinned-original-source.tex, source lines 476–479. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $f(x_0, \ldots, x_k)$ is representable in* ThQ*, there is formula $A(x_0, \ldots, x_k, y)$ such that*

$$\text{Th}Q \vdash A_f(\overline{n_0}, \ldots, \overline{n_k}, \overline{m}) \quad \textit{iff} \quad m = f(n_0, \ldots, n_k).$$

Proof. The "if" part is ????. The "only if" part is seen as follows: Suppose Th$Q \vdash A_f(\overline{n_0}, \ldots, \overline{n_k}, \overline{m})$ but $m \neq f(n_0, \ldots, n_k)$. Let $l = f(n_0, \ldots, n_k)$. By ????, Th$Q \vdash A_f(\overline{n_0}, \ldots, \overline{n_k}, \overline{l})$. By ????, $\forall y\,(A_f(\overline{n_0}, \ldots, \overline{n_k}, y) \to \overline{l} = y)$. Using logic and the assumption that Th$Q \vdash A_f(\overline{n_0}, \ldots, \overline{n_k}, \overline{m})$, we get that Th$Q \vdash \overline{l} = \overline{m}$. On the other hand, by lem-q-proves-neq, Th$Q \vdash \overline{l} \neq \overline{m}$. So Th$Q$ is inconsistent. But that is impossible, since ThQ is satisfied by the standard model (see ??), $\mathfrak{N} \vDash$ ThQ, and satisfiable theories are always consistent by the Soundness Theorem (prfAX/fol:axd:sou:cor:consistency-soundness, prfSC/fol:seq:sou:cor:consistency-soundness, prfND/fol:ntd:sou:cor:consistency-soundness, prfTab/fol:tab:sou:cor:consistency-soundness). □